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DJ Algorithm Implementation

Video: Implementing the Deutsch-Jozsa Algorithm: Oracle Functions in Qiskit
6 min

Video: Building and Running a Deutsch-Jozsa Quantum Circuit in Qiskit
6 min

Video: Running the Deutsch-Jozsa Algorithm: Constant vs Balanced Oracles
3 min

Graded Assignment: Assessment 14
Submitted

Assessment 14

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

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Assignment details

Due Dec 27, 11:59 PM IST

Attempts Unlimited

Submitted Dec 11, 3:51 AM IST

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92%

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Next item →

1. In the context of quantum computing, what does the term 'superposition' refer to? **1 / 1 point**

- ○ The probability of a qubit being measured in the $|0\rangle$ state.
- ◉ A qubit being in a state that is a combination of $|0\rangle$ and $|1\rangle$.
- ○ A qubit being in either $|0\rangle$ or $|1\rangle$, but not both.
- ○ The interference pattern created by multiple qubits.

Correct
Correct! Superposition allows a qubit to be in a combination of both states simultaneously.

2. Which of the following operations are used to create superpositions in quantum computing? **1 / 1 point**

- ☑ Quantum Fourier Transform

  Correct
  Correct! The Quantum Fourier Transform can also be used to create superposition states.

- ☑ Hadamard gate

  Correct
  Correct! The Hadamard gate is commonly used to create superpositions.

- ☐ Pauli-X gate
- ☐ Pauli-Z gate

3. Which of the following steps are involved in designing a balanced Oracle circuit in Qiskit? **0.8 / 1 point**

- ☐ Apply an X gate to all qubits to create superposition.
- ☑ Initialize the quantum circuit and define input qubits.

  Correct
  Correct! Initializing the circuit and defining input qubits are essential steps in designing a balanced Oracle circuit.

- ☐ Apply a sequence of Hadamard gates to create a balanced state.
- ☐ Measure the qubits to get the output probabilities.
- ☑ Define the Oracle function to map specific inputs to outputs.

  Correct
  Correct! Defining the Oracle function is a key step in designing a balanced Oracle circuit.

You didn't select all the correct answers

4. What is the purpose of plotting a histogram of the results after executing a quantum circuit multiple times? **1 / 1 point**

- ○ To identify errors in the quantum gates used.
- ◉ To visualize the probabilities of different outcomes.
- ○ To compare the performance of different quantum algorithms.
- ○ To determine the exact state of each qubit.

Correct
Correct! Plotting a histogram helps visualize the probabilities of different outcomes, providing insight into the quantum state's distribution.

5. Which of the following are components of implementing a constant Oracle function in Qiskit? **0.8 / 1 point**

- ☐ Initializing qubits to a superposition state using Hadamard gates.
- ☑ Defining the Oracle function that produces a constant output for all inputs.

  Correct
  Correct! The essence of a constant Oracle function is to produce a constant output, regardless of the input.

- ☐ Applying a series of controlled NOT gates based on the Oracle function.
- ☑ Using a quantum register to define qubits.

  Correct
  Correct! A quantum register is necessary to define the qubits involved in the Oracle function.

- ☐ Creating a classical register to store measurement results.

You didn't select all the correct answers